SOFTWARE REQUIREMENTS SPECIFICATION

KEYSPRINT

A browser-based competitive typing-race platform

Concept inspired by TypeRush; distinctive branding, interface, game artwork, and architecture.

Document	Specification
Version	1.0 - Product baseline
Date	27 October 2026
Status	Draft for product and engineering approval
Platform	Responsive web application (desktop-first; mobile supported)
Reference	https://www.typerush.com/
Project name	KeySprint - placeholder brand, subject to approval

Prepared for product owner, UI/UX, frontend, backend, game engineering, QA and DevOps teams.

Document control

Item	Definition
Purpose	State verifiable product, user, design, data, integration, security and operational requirements.
Priority definitions	MUST = required for stated release; SHOULD = high-value, may be deferred; COULD = optional extension.
Release convention	MVP = launch baseline; V1 = complete core product; V2 = enhancements.
Change management	Any changes to scoring, anti-cheat rules, rewards, age handling or multiplayer synchronization require a versioned specification update.
Primary assumptions	English keyboard typing at launch; virtual currency has no cash value; browser-only product; suggested working brand.

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1. Executive overview and reference analysis

KeySprint is a skill-based multiplayer web game that turns typing practice into an animated race. The player types an assigned passage accurately; correct sequential progress moves their vehicle forward on a visual track. Players can enter public races, join friends in private rooms, race computer-driven opponents, practice alone, earn non-redeemable game currency, unlock cosmetic vehicles, and track improvement.

1.1 Reference features observed

Public TypeRush pages describe typing races against other players or AI opponents, race-based money and XP rewards, unlockable/customizable cars and boats, worldwide/country ranking, friend invitations, achievements, race history, statistics, and personalized difficult-key exercises. The exact interactive game behavior could not be completely verified through static public pages; mechanics below are proposed requirements for KeySprint, not claims that TypeRush implements the same rules.

Reference capability	KeySprint direction
Typing-powered races	Keep the same core skill mechanic; adopt an original 2D arcade presentation.
Multiplayer / AI opposition	Use real-time server-authoritative rooms and configurable bots.
Money, XP, collectible vehicles	Use fair-play-only cosmetic rewards; no performance advantage for paid or premium items.
Country/global rankings	Add period filters, verified runs and anti-cheat review.
Friends and invitation rewards	Create sharable private rooms; add friend networks and referral rewards in V1.
Typing analytics	Offer readable WPM, accuracy, mistakes and key-specific practice.

1.2 Differentiation

- Visual direction: a clean, modern arcade interface (dark navy or soft off-white foundations, electric teal/cyan highlights, high-contrast type, original illustrations).
- Race presentation: side-on multi-lane track, smooth progress animation, status indicators, and optional selectable track themes.
- Practice experience: instant training without registration, per-key insights, and accessibility-friendly motion controls.
- Competitive fairness: deterministic passages, matched lengths, server validation, and cosmetics only.
- Extensibility: separate race engine, profile/reward services, moderation tools, and structured telemetry.

2. Objectives, personas and scope

2.1 Business and player objectives

- Make skill improvement enjoyable and measurable while encouraging repeat visits.
- Let a new visitor begin solo practice or a quick race with no mandatory sign-up.
- Offer fair real-time competition for friends and globally matched players.
- Build retention through user progress, achievements, visual collectibles and seasonal events.
- Provide administrators control over content, reward policy, game availability and moderation.

2.2 User roles

Role	Permissions / needs
Visitor / Guest	Play solo and selected public races; view public leaderboards; optional local transient stats.
Registered Player	Persistent profile, race history, leaderboard eligibility, rewards, garage, friend invites.
Moderator	Review reports, hide nicknames, suspend cheating/abusive accounts, inspect flagged races.
Administrator	Manage content, vehicles, race parameters, events, players, metrics and staff rights.
System / Bot	Generate bot progress events for non-human racers under deterministic profiles.

2.3 Release scope

Area	MVP	V1	V2
Gameplay	Solo + public real-time + bot fallback	Private rooms, friends, daily challenges	Ranked seasons, tournaments
Accounts	Guest + email/Google sign-in, recovery	Profile/privacy refinements	School or team accounts
Rewards	XP, coin wallet, basic garage	Achievements, advanced cosmetics	Seasonal collections
Stats	WPM/accuracy/history	Difficult-key drills and trend charts	Advanced learning plans
Admin	Users, passages, race health	Achievements, reports, inventories	Event/season operations
Localization	English content and keyboard	Additional language packs	Region-specific competitions

3. User journeys

3.1 First-time visitor

1. Open landing page and see prominent Play Now / Practice buttons.
2. Choose guest name and avatar (optional), then choose Quick Race or Practice.
3. For Quick Race: enter queue, see loading and countdown, then type the shared passage.
4. See lane movement, current WPM, accuracy, mistakes and finish position in real time.
5. On result page, view score and prompt to create an account to save progress.

3.2 Registered public race

6. Sign in and load preferred vehicle, country league and prior skill rating.
7. Enter matchmaking; show players in room, wait status and countdown.
8. Type same passage as all competitors; movement reflects validated correct progress.
9. On finish or timeout, server freezes ranking and publishes trustworthy results.
10. Reward service credits XP/currency once, stores log and updates eligible rankings.

3.3 Challenge a friend

11. Player creates a private room and receives shareable URL / room code.
12. Invited users join lobby; host selects difficulty and start option.
13. Room starts when host activates countdown, with bot slots optional.
14. Players finish and compare results; room offers Rematch and Copy Invite.

3.4 Practice and progression

15. User selects timed, passage or difficult-key mode.
16. Practice UI displays live correctness and speed without race opponents.
17. Session summary identifies error-prone keys, speed and consistency.
18. Player can retry, select a practice recommendation, or switch into competition.

4. Functional requirements

4.1 Public website and navigation

FR-W01 **Landing page** **[MUST]** Show product explanation, animated gameplay preview, Start Race, Practice, leaderboards, feature highlights and authentication entry.

FR-W02 **Navigation** **[MUST]** Persistent or mobile-adapted navigation for Play, Practice, Leaderboards, Garage, Achievements, Friends, Profile and Support.

FR-W03 **Public content** **[MUST]** Provide About, How to Play, Rules, Privacy, Terms and Contact pages; avoid claims of real-money earnings.

FR-W04 **SEO** **[MUST]** Index marketing and help pages; keep live race, accounts and private lobbies out of public indexing.

4.2 Registration, login and profile

FR-A01 **Guest entry** **[MUST]** Allow one-click guest sessions with a nickname; clearly disclose transient data handling.

FR-A02 **Authentication** **[MUST]** Support email/password and Google OAuth or OIDC; verify email for persistent and competitive features.

FR-A03 **Profile** **[MUST]** Allow display nickname, unique ID, selected avatar/vehicle, verified country, skill statistics and privacy settings.

FR-A04 **Recovery and sessions** **[MUST]** Support password reset, logout across sessions, account deletion and secure refresh/session expiry.

FR-A05 **Guest conversion** **[MUST]** On registration, merge permitted guest session scores only after validation; never allow duplicate reward claims.

FR-A06 **Age and consent** **[MUST]** Apply an age-appropriate access policy, parental consent where legally required, and limited-data child-safe mode as applicable.

4.3 Race modes

FR-R01 **Quick Race** **[MUST]** Provide public matchmaking with 2-6 racers per room, configurable by administrator; bots fill open positions if needed.

FR-R02 **Practice Race** **[MUST]** Offer unlimited solo practice, non-competitive by default and with no ranked score credit.

FR-R03 **Bot Race** **[MUST]** Make multiple bot difficulty profiles available: Easy, Normal and Advanced; clearly label bots.

FR-R04 **Private Room** **[SHOULD]** Host generates a protected share code, chooses room capacity and shares an invite URL.

FR-R05 **Friend Race** **[SHOULD]** Invite connected friends, accept invitations, and show Rematch and Return to Lobby.

FR-R06 **Modes extension** **[COULD]** Prepare game engine for tournaments, country events, teams, leagues and timed challenge races.

4.4 Race engine and live user interface

FR-G01 **Shared passage** **[MUST]** All human racers in a room receive the same passage/version, punctuation, language and normalization policy.

FR-G02 **Countdown** **[MUST]** Show lobby status, synchronized 3-second countdown and authoritative server start timestamp.

FR-G03 **Typing capture** **[MUST]** Provide focused typing input with correct/incorrect highlighting, backspace handling and explicit paste prevention.

FR-G04 **Vehicle movement** **[MUST]** Render distance proportional to validated correctly completed passage progress; cosmetics do not change speed.

FR-G05 **Live HUD** **[MUST]** Display elapsed time, personal live WPM, accuracy, progress percentage, opponents and finish order.

FR-G06 **Race completion** **[MUST]** Finish when entire passage is correctly committed or server timeout occurs; show authoritative ranking.

FR-G07 **Room resilience** **[MUST]** Support reconnect within configurable grace period, clear paused/disconnected status and deterministic quit handling.

FR-G08 **Audio and motion** **[MUST]** Offer independent toggles for audio, effects and reduced motion; never make sound essential to performance.

FR-G09 **Fair competition** **[MUST]** Validate monotonic progress, reasonable input timing and final results server-side; suspicious scores enter review.

FR-G10 **Accessibility** **[MUST]** Support keyboard-only navigation and screen-reader friendly practice mode; provide static/low-motion race view.

4.5 Scoring, reward wallet and XP

FR-S01 **Measurements** **[MUST]** Calculate WPM, accuracy, duration, corrected errors, final progress, place and optional consistency with versioned formulas.

FR-S02 **Race results** **[MUST]** Create immutable result record for every completed match, including timestamp, mode, passage ID, ranked status and rule version.

FR-S03 **Reward policy** **[MUST]** Award non-cash in-game credits, XP and league points from configurable rules based on completion, accuracy and placement.

FR-S04 **Wallet ledger** **[MUST]** Store credits/debits as auditable transactions; prevent negative balances and duplicate race reward credit.

FR-S05 **XP and level** **[MUST]** Determine account level from cumulative XP using a centrally managed progression table.

FR-S06 **Unrated results** **[MUST]** Give zero league points for abandoned, bot-only, tampered or otherwise non-eligible races.

4.6 Garage, store and customizations

FR-C01 **Garage** **[MUST]** Show available, owned, selected and locked vehicles with metadata and preview.

FR-C02 **Store** **[MUST]** Allow virtual currency purchase of cosmetic vehicles, paints, tires/trails and track decorations.

FR-C03 **Entitlements** **[MUST]** Server validates balance and purchase prerequisites before granting items; purchase transaction must be atomic.

FR-C04 **Equal speed** **[MUST]** Every cosmetic option must be strictly non-pay-to-win and provide identical progression physics.

FR-C05 **Multiple vehicle groups** **[SHOULD]** Design data model for cars at MVP; hoverboards/boats or other themed fleets can be added later.

4.7 Leaderboards and competitive rankings

FR-L01 **Global ranking** **[MUST]** Show top performers by verified best WPM, league points and race count with tie-break logic.

FR-L02 **Country rankings** **[MUST]** Filter by user-selected eligible country with verification and change-limit policy.

FR-L03 **Periods** **[SHOULD]** Support all-time and weekly periods at MVP; monthly, season and friend-based filters at V1.

FR-L04 **Rank visibility** **[MUST]** Show personal position even when outside current displayed page; paginate or virtualize long results.

FR-L05 **Integrity** **[MUST]** Exclude banned, flagged and unverified races from public records until accepted.

4.8 Friends, invitations and social

FR-F01 **Friend list** **[SHOULD]** Search users with consent/privacy controls, send/accept/delete friend requests.

FR-F02 **Room invitations** **[SHOULD]** Copy private room link and send through installed share tools; do not demand contact permissions.

FR-F03 **Privacy controls** **[SHOULD]** Allow invite blocking, public profile visibility setting and ability to report users.

FR-F04 **Referral bonuses** **[COULD]** Optional capped bonuses after a unique qualified user completes a legitimate first race.

4.9 Achievements and missions

FR-H01 **Achievement rules** **[SHOULD]** Support milestones for races completed, speed, accuracy, streak and total progress.

FR-H02 **Badge award** **[SHOULD]** Grant badges once with idempotent event processing and visible requirement/progress.

FR-H03 **Daily challenges** **[SHOULD]** Offer rotating tasks, reward cap and UTC/local display clarity.

4.10 Typing training and analytics

FR-T01 **Modes** **[MUST]** Offer passage practice, timed tests (15/30/60/120 seconds), numbers and punctuation.

FR-T02 **Personal analytics** **[SHOULD]** Graph 7-day and 30-day performance; show best and median/average verified WPM, accuracy and practice time.

FR-T03 **Weak keys** **[SHOULD]** Track per-key error counts where consent/policy permits; recommend targeted drills.

FR-T04 **Content levels** **[MUST]** Use beginner/intermediate/advanced passage categories with curated difficulty labels.

4.11 Notifications and support

FR-N01 **In-app feedback** **[MUST]** Show unobtrusive join, disconnect, reward, badge and invite notifications.

FR-N02 **Support and reports** **[MUST]** Provide bug report, offensive nickname report, cheating report and contact/help workflows.

FR-N03 **Transactional email** **[MUST]** Send verification, password recovery and opt-in invitation notifications via a managed provider.

4.12 Administrative console

FR-D01 **Dashboard** **[MUST]** Display DAU, sessions, races, completion rate, WPM distribution, queue health and runtime errors.

FR-D02 **Player management** **[MUST]** Search, view, suspend, restore, deactivate, anonymize, and export eligible player records with staff audit trail.

FR-D03 **Passage management** **[MUST]** Create/edit/import/retire text passages; version and approve before publication.

FR-D04 **Game configuration** **[MUST]** Edit countdown, room size, bots, timeout, XP/reward formula, event toggles and basic feature flags.

FR-D05 **Catalog and achievements** **[SHOULD]** CRUD vehicle SKUs, cosmetics, unlock criteria and badge rules.

FR-D06 **Reports and review** **[SHOULD]** Triage flagged sessions, analyze typing timings and invalidate unfair outcomes.

FR-D07 **Admin roles** **[MUST]** Restrict dangerous functions with role-based access control and record actor/action/time/old-new value.

5. Game mechanics and business rules

5.1 Authoritative calculations

For a text passage consisting of N required characters, only sequential, correct, committed characters count toward track progress. Client live figures are previews; server final figures are the source of truth.

Metric	Required formula / policy
Race progress	validated correct committed characters / target characters * 100, capped to [0,100].
Gross WPM	total submitted typed characters / 5 / elapsed minutes; informational only.
Net WPM	correct validated characters / 5 / elapsed minutes; used as the displayed final WPM baseline.
Accuracy	correct submitted keystrokes / total submitted printable keystrokes * 100; backspaces are counted separately.
Errors	count incorrect printable input attempts; corrected errors tracked independently from final remaining errors.
Finish placement	ascending server-recorded finish timestamp; ties resolved by greater accuracy, then stable participant ID.
Duration	server_finish_time - server_start_time, excluding no post-finish idle time.
Ranked eligibility	registered/verified player + approved room + completed race + passing anti-cheat checks.

Note: the exact accuracy policy must be confirmed at implementation because backspaces and composition input can be measured in multiple ways. Tests must use the same versioned policy across web client, API, leaderboard and reporting.

5.2 Match lifecycle

States: CREATED -> WAITING -> COUNTDOWN -> RUNNING -> COMPLETED / TIMED_OUT / CANCELLED. At each transition, the server emits a monotonic sequence number and rejects out-of-order state changes.

- Suggested default: 2-6 competitors and one passage of about 180-300 English characters (administrator configurable).
- Public queue should seek similarly skilled opponents for a configurable 8-second target; after a maximum wait (proposed 12 seconds), fill with clearly labeled bots.
- Countdown is synchronized; no scoring before server start; late joiners cannot enter a race already RUNNING.
- Suggested maximum race duration 150 seconds; unfinished players are ranked by validated progress, then accuracy at timeout.
- If a client disconnects, continue authoritative timer and allow an approximately 15-second reconnection grace interval; abandon when time runs out.
- Race completion, wallet payout and ranking eligibility are each processed exactly once for a given race/player.

5.3 Proposed reward examples (configurable)

Action	XP	Credits	League points
Valid completed public race	50 base + place bonus	100 base + accuracy bonus	Based on placement and verified speed
Practice session	10 (daily capped)	0	0
Private race	25 (daily capped)	0	0
Flagged / invalid result	0	0	0

Values above are illustrative configuration defaults, not mandatory final economy amounts. All amounts are virtual, non-transferable, non-redeemable and have no cash or monetary value.

5.4 Anti-cheat checks

- Block paste/drop into competitive typing input; capture event timestamps with a server-side validation policy.
- Use progressive anomaly detection for unrealistic typing spikes, identical timing sequences, automation, duplicate accounts and improbable error-free speed.
- Do not expose full unpublished passage pool before match; rotate or cache-protect content appropriately.
- Never trust client-supplied WPM, XP, place, wallet balance or leaderboard values.
- Quarantine suspicious records instead of automatically publicizing high scores; support moderator appeals and auditability.

6. Screens and design guidelines

6.1 Screen inventory

Screen / route	Purpose and required content	Phase
/	Marketing landing, game preview, calls to action	MVP
/play	Mode selection and match entry	MVP
/queue	Matchmaking / bots / cancel	MVP

/race/:id	Live lanes, passage, typing area, HUD, status	MVP
/results/:id	Placement, WPM, accuracy, rewards, rematch	MVP
/practice	Mode selection, drills, typing test	MVP
/login, /register, /reset	Authentication and guest conversion	MVP
/profile/:id	Account card, achievements, preferred vehicle	MVP
/stats	History, accuracy, trends, weak keys	V1
/garage, /store	Catalog, currency balance, cosmetics	MVP
/leaderboards	Global/country/weekly leaderboard	MVP
/friends	Friends, friend requests and invites	V1
/rooms/:code	Private lobby, invite, start and rematch	V1
/achievements	Unlocked badges, progress and rewards	V1
/settings	Sound, motion, keyboard, theme, privacy	MVP
/help, /terms, /privacy	Guides, support and legal policies	MVP
/admin/*	Protected operations console	MVP/V1

6.2 Visual direction and UI behavior

- Avoid copying TypeRush layouts, logos, named vehicle designs, proprietary artwork, text passages or UI assets.
- Visual theme: original modern arcade game; deep navy #102237, cyan/teal #00BFA6, sky blue #3E88E8, warm accent #FFC857; white/light alternate theme.
- Typography: clear sans-serif UI, optional monospace practice text; prioritize legibility over decoration.
- Home: hero illustration + one obvious Play Now action, live mini-leaderboard and accessible explanation.
- Race: two-column desktop layout with track above typing zone or track left/text right; six compact lanes maximum.
- Mobile: stack race track, live statistics and typing area; account for on-screen keyboard viewport.
- Feedback: valid typed letters, incorrect letters, current target, speed changes, finish ranking, and earned rewards visually differentiated.
- Animations: 60 fps target where supported, graceful static fallback, no flashing/high-intensity effects.
- Components: reusable cards, controls, avatar chips, vehicle previews, rankings, modals, notifications and progress bars.

6.3 Responsive breakpoints

Device	Behavior
>= 1200 px	Full lane track, expanded stats, rich leaderboard and profile panels.
768-1199 px	Reduced lane widths and compact navigation, no critical clipping.
< 768 px	Stacked gameplay, touch-friendly controls, optional soft keyboard, simplified animation.

Low-performance / reduced motion	Minimal graphics and optional text-focused mode; same scoring.

7. Architecture, data and APIs

7.1 Suggested system architecture

Recommended stack: Next.js (React + TypeScript) for marketing/app routes; Canvas or PixiJS for high-performance 2D racing (plain CSS/Canvas acceptable at MVP); NestJS/Node.js REST API with Socket.IO or native WebSocket gateway; PostgreSQL for accounts, matches and ledger; Redis for queues, short-lived rooms and rate limiting; S3-compatible object storage for original artwork/assets; CDN and managed email/OAuth providers. This is a recommended stack, not a mandatory vendor lock-in.

Request flow: Browser UI -> REST for persistence/metadata -> WebSocket gateway for matchmaking and live match state -> server-authoritative race engine -> PostgreSQL transaction for results/rewards -> leaderboard read model/cache. Redis is ephemeral, never the only source of reward balances.

7.2 Core data model

Entity	Key fields / relationships
users	id, email, auth_provider, status, created_at, verified_at, age_policy_state
profiles	user_id, nickname, avatar_id, country_code, privacy_flags
player_stats	user_id, verified_best_wpm, avg_wpm, accuracy, races, xp, level
passages	id, locale, text, text_hash, difficulty, status, version
race_rooms	id, mode, status, passage_id, start_at, timeout_at, region
race_participants	id, race_id, user_or_guest_id, vehicle_id, bot_flag, join_at, finish_at
race_input_events	race_id, participant_id, ordered event/timing summary or secure telemetry reference
race_results	id, race_id, participant_id, duration, wpm, accuracy, place, eligibility, rule_version
wallets and wallet_transactions	user_id, balance projection, delta, reason, idempotency_key, race_result_id
vehicles / cosmetics	id, category, price, unlock_rules, active, asset_url
inventory	user_id, catalog_item_id, purchased_at, selected_at
friendships / invites	requester_id, receiver_id, status, invite_token, expiry
achievements / user_achievements	criteria JSON, earned_at, reward_transaction_id
leaderboard_snapshots	period, country, user_id, verified_metric, ranking
moderation_reports / audit_logs	target, reason, review status, actor, timestamp, action payload

7.3 REST and real-time interface sketch

Method / channel	Purpose
POST /api/v1/auth/*	Sign up, login, refresh, logout, password reset
GET/PATCH /api/v1/me	Profile and settings read/write
POST /api/v1/matchmaking/join	Create or join public queue
POST /api/v1/rooms	Create private room and room code
GET /api/v1/races/:id	Room and result snapshot
GET /api/v1/leaderboards	Metric, country, period and pagination
GET /api/v1/garage; POST /api/v1/store/purchase	Inventory and cosmetic purchase
GET /api/v1/stats; GET /api/v1/achievements	Stats and progression
GET/POST /api/v1/admin/*	Authorized admin operations and reviews
WS: match.join / match.ready / match.start	Match lifecycle events
WS: input.batch / progress.update	Bounded client input / server-validated progress
WS: match.finish / match.abort / match.reconnect	Server result and resilience events

7.4 Real-time transport contract

- All socket messages include protocol version, room ID, event ID, monotonic sequence and server timestamp where applicable.
- Authenticate socket handshakes, revalidate permissions on room join and rate-limit high-frequency input messages.
- Batch or throttle telemetry; do not send one database write per keystroke.
- Server sends compact state snapshots and incremental updates; reconnect path requests a consistent authoritative snapshot.
- Persist match outcomes transactionally after completion; message retries cannot duplicate rewards.

8. Non-functional requirements

ID	Requirement	Acceptance target
NFR-01	Availability	Target 99.5% monthly availability at launch, excluding announced maintenance.
NFR-02	Web responsiveness	Marketing pages LCP <= 2.5 sec at p75 on supported broadband devices (target).
NFR-03	Realtime latency	Regional match progress update p95 <= 300 ms after accepted message under load-test network conditions.
NFR-04	Concurrency	Demonstrate 500 concurrent active racers in staging before MVP go-live; scale target thereafter.
NFR-05	Race recovery	Reconnected user recovers authoritative room snapshot within 3 sec at p95 after handshake.

NFR-06	Client performance	Keep rendering responsive on modern desktop; degrade effects on slow devices.
NFR-07	Browser support	Latest two major Chrome/Edge/Firefox/Safari versions on desktop and current mobile browsers.
NFR-08	Accessibility	Aim for WCAG 2.2 AA across general navigation, practice and result screens; document unavoidable game-specific limits.
NFR-09	Observability	Structured events, errors, trace IDs, queue metrics, alerts and dashboards.
NFR-10	Backups	Daily database backups with point-in-time recovery where available; periodic restore drills.
NFR-11	Data durability	Wallet credit/purchase and race result writes must be atomic and idempotent.
NFR-12	Localization	i18n-ready UI strings and content; English only is minimum launch content.

9. Security, privacy and moderation

9.1 Security controls

- HTTPS and secure cookies; protect against CSRF as relevant, XSS and SQL injection; never trust unvalidated input.
- Strong password hashing (Argon2id or equivalent), OAuth PKCE where applicable, token rotation, session revocation.
- RBAC, staff MFA, protected administrative routes and immutable audit trail for sensitive changes.
- Rate limits and bot defenses for signup, login, queueing, invite, chat-equivalent interactions and public APIs.
- Validate public nicknames for abuse, character length and impersonation; provide reporting/moderator tooling.
- Use random, high-entropy room codes/links with TTL and optional host removal controls.
- Use secure CI/CD, secret storage, dependency scanning, backups and infrastructure access separation.

9.2 Data and minors

- Publish explicit privacy policy, retention schedules, deletion/export requests and cookie notices as applicable.
- Avoid collecting unnecessary sensitive personal data or raw text the user never intended to submit.
- Typing timing signals used to detect cheating can be behavioral data; document purpose, retention and access scope.
- Define minimum age and parental consent/supervision requirements with counsel for each launch market; underage player visibility defaults to conservative settings.
- Public leaderboards should show nickname and voluntary country, not private email, birth date or identifying details.
- The default product has no cash gambling, cash rewards, token purchases or real-world financial exchange.

10. Administration, analytics and operations

10.1 Admin work areas

- Overview: activity, active rooms, queue wait, abandoned races, suspicious-result rate and uptime.
- Users & moderation: search player, sanctions, reports, block lists and appeals.
- Content CMS: add passages, language, difficulty, status, duplicate detection and text length checks.
- Economy: catalog, reward rules, wallet corrections with required reason and dual approval for large corrections.
- Competitive: rankings review, suspicious scores, seasonal reset and event schedule (later phase).
- System: feature flags, operational alerts, maintenance mode, incident notes and staff audit logs.

10.2 Product analytics events

Event	Important properties
landing_view, play_click	anonymous/session ID, source, device category
queue_join, queue_match, queue_cancel	mode, wait_ms, region, bot_count
race_start, race_finish, race_abandon	race_id, participant count, elapsed_ms, finish/place, WPM, accuracy
signup_start, signup_success	source, anonymous conversion flag
practice_start, practice_complete	mode, duration, speed, error category
garage_view, purchase_success	catalog item ID, virtual price, resulting balance
friend_invite, private_room_start	invite method, room count; no private contact data in analytics
anti_cheat_flag, score_rejected	rule_id, confidence class, moderator disposition

10.3 Proposed KPIs

- Visitor -> first race conversion; visitor -> registered account conversion.
- Average queue time, start rate, completion rate, multiplayer participation percentage.
- DAU/WAU, D1/D7 retention, median races per user, practice minutes.
- Verified median/upper percentile WPM trend and accuracy improvement.
- Crash-free sessions, suspicious-score review load, admin moderation turnaround.

11. Acceptance criteria and testing

11.1 Critical user acceptance tests (for applicable release)

AT-01 **Test** **[MUST]** Guest starts solo practice without registering and receives WPM/accuracy results.

AT-02 **Test** **[MUST]** Two racers in same room see identical passage version and synchronized countdown.

AT-03 **Test** **[MUST]** Vehicle cannot advance through incorrect unresolved characters.

AT-04 **Test** **[MUST]** Race finish ranking is computed once from server state, not client timestamp.

AT-05 **Test** **[MUST]** Paste attempt does not advance a ranked race.

AT-06 **Test** **[MUST]** A disconnected player reconnects to the same room within grace period with consistent progress.

AT-07 **Test** **[MUST]** A timed-out racer has correct partial-progress placement and no unauthorized rewards.

AT-08 **Test** **[MUST]** Exactly one reward credit is issued even after duplicate result messages.

AT-09 **Test** **[MUST]** Wallet purchase fails if funds are insufficient, with no partial entitlement.

AT-10 **Test** **[MUST]** Any available cosmetic vehicle yields identical position at identical typing progress.

AT-11 **Test** **[MUST]** Unverified/flagged account is excluded from eligible rankings.

AT-12 **Test** **[MUST]** A newly registered player can recover an account using verified email.

AT-13 **Test** **[MUST]** Private room rejects expired code and prohibits joining started games.

AT-14 **Test** **[MUST]** Country/global board sorting and filtering return expected deterministic order.

AT-15 **Test** **[MUST]** Moderation action is logged and only permitted staff can enact it.

AT-16 **Test** **[MUST]** Admin can retire a passage without affecting historical race result integrity.

AT-17 **Test** **[MUST]** Keyboard-only user can enter practice mode, submit test and reach results.

AT-18 **Test** **[MUST]** Mobile virtual keyboard does not cover the active typing text and progression.

AT-19 **Test** **[MUST]** A suspicious 200+ WPM burst is flagged according to server-configured thresholds, not automatically accepted.

AT-20 **Test** **[MUST]** A player who requests account deletion receives a policy-compliant resolution without corrupting shared race history.

11.2 Quality gates

- Automated unit tests on scoring rules and event order, including Unicode, punctuation, backspace, IME policy and tie-breaking.
- Integration tests for queue -> room -> race -> result -> ledger -> leaderboard with failure/retry cases.
- Load and soak tests for WebSocket rooms, concurrent participants and reconnect storms.
- Cross-browser and mobile QA for input focus, keyboard layout, canvas performance and responsive design.
- Security testing for API authorization, malicious socket input, abusive nicknames, referral farming and economy exploits.
- Usability tests with new typists and experienced players; verify clarity of accuracy, WPM and navigation.

12. Delivery plan and exclusions

12.1 Recommended implementation phases

Phase	Indicative workstream / output	Gate
0 - Discovery / UI	Finalize brand, game rules, privacy policy, wireframes and visual design system.	SRS approved + screen designs signed off
1 - Core engine	Auth/guests, passage editor, solo practice, server race state, WebSocket, bots, race results.	Reliable end-to-end functional race
2 - MVP production	Public matchmaking, scores, basic garage, rewards, leaderboards, admin, deployment, testing.	Beta + acceptance tests passed
3 - V1 social	Private rooms, friends, badges, charts, difficult-key training, improved moderation.	Social and learning release
4 - V2 growth	Tournaments, leagues/seasons, language packs, school/team features, advanced customization.	Roadmap after telemetry review

Effort, timeline and budget should be estimated after final branding, concurrency target, artwork quality, mobile-input policy and MVP/V1 scope are approved. This document deliberately does not assume a fixed budget or delivery date.

12.2 Explicit exclusions from MVP

- Native Android or iOS apps; web is the delivery target.
- Real-money prizes, crypto, betting, gambling, cash-out or marketplace exchanges.
- Live text chat, user-generated race passages and public voice chat.
- Premium purchase checkout and subscriptions, unless separately approved.
- Full school administration, classroom moderation and teacher analytics.
- Complex 3D vehicle simulation, 3D track editor and open-world racing.

12.3 Key dependencies and risks

Risk	Mitigation
Competitive latency and desync	Server-authoritative game loop, regional gateway, reconnect snapshots.
Cheating inflates leaderboards	Validation policy, suspicious-score quarantine, audit log and appeals.
Keyboard inconsistency on mobile	Mobile input prototype before final gameplay UI sign-off.
Economy becomes exploitable	Idempotent ledger, item purchase transactions and reward caps.
Licensing of art and passages	Create original assets and licensed/public-domain or commissioned typing content.
Scope expansion	Separate MVP, V1, V2 with change-control process.

13. Reference materials and open decisions

13.1 Public reference pages reviewed

- TypeRush overview: https://www.typerush.com/
- TypeRush achievements: https://www.typerush.com/achievements.html
- TypeRush garage: https://www.typerush.com/garage.html
- TypeRush friends: https://www.typerush.com/friends.html
- TypeRush results: https://www.typerush.com/results.html
- TypeRush global leaderboard: https://www.typerush.com/global-leaderboard.html
- TypeRush sign-up: https://www.typerush.com/sign-up.html

13.2 Product-owner sign-off questions

- Final brand/domain and preferred style (neon-dark, light-minimal or sport/arcade).
- Launch geography, supported languages and country league verification policy.
- Mobile typing: on-screen keyboard fully supported, or keyboard-connected play encouraged.
- Which features are contractually included in MVP versus V1; confirm private rooms in launch or follow-up.
- Max users and simultaneous races for initial infrastructure sizing.
- Virtual-currency/reward amounts, rules, unlock prices, catalog size and target achievement count.
- Moderation roles, minimum age and legal/privacy policies by market.
- Artwork scope (2D cars, hoverboards/boats, map themes) and audio production.

13.3 Approval record

The following are sign-off placeholders; document approval is not implied by distribution.

Role	Name / signature	Date
Product Owner	_____	_____
Engineering Lead	_____	_____
UI/UX Lead	_____	_____
QA Lead	_____	_____

END OF SPECIFICATION • VERSION 1.0